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DSE501 : Introduction to Data Science Tools
Prepared by : Prof. Dr. Hasan DAĞ
Date Assigned : 26.10.2026
Date Due : 02.11.2026 by midnight (23:45)
Exam Type : **Hwk 1b**

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Pre-homework chore: Learn struct data type from the **PowerPoint slides I shared on learn.khas.edu.tr**, or from the **w3schools.com**, or from the videos, for which I share a couple of links in below.

A) https://www.youtube.com/watch?v=dqa0KMSMx2w&t=10s
B) https://www.youtube.com/watch?v=e9OXqEPF-5M
C) https://www.youtube.com/watch?v=IAvfAC4H_0s

Homework Problem Statement (*You just do the same homework1a with struct data type***):** Write a program that reads personal data and address data for the instructors from a data file (max. 50 records, data belonging to instructors) named as ***DSE501_Hwk1*_Instructors.txt** and *inserts them into a data structure made up of* ***struct data type-based array in a sorted manner*** *with respect to the age. Then print the whole data to the screen. Reading data from a file line by line, breaking every line into pieces, inserting the pieces to the relevant data structure, sorting the data, and finally printing the data shall all be done by separate functions. That is, your main program is expected as short as possible. To this end, you are required to develop a library file (.h file) called* ***NameLastname_DSE501_Hwkb1_Library.h*** *as well.*

Please name your Homework1b code file as ***NameLastname_ DSE501_Hwk1b.c*** *(failure to do so will cause a deduction of 5 points) and upload both the library the file and your homework file as a single zipped file to* ***learn.khas.edu.tr***

The content of a sample ***DSE501_Hwk1*_Instructors.txt**
(Name, Last-name, Age, Gender, Street Name, Street Number, City, Country)

```
Hasan, Dag, 61, M, Cibali, 1, Istanbul, Turkey.
Reiner, Creuztburg, 71, M, Ernst-Reuter Platz, 10, Berlin, Germany.
Jane, Schwarz, 34, F, Jung Street, 23, Chicago, USA.
…
… (more lines, total will be <= 50 lines)
..
Klaus, Tomy, 21, M, Blume Street, 98, Potsdam, Germany.
```

Notes:

1. This homework requires a good understanding of the topics below.
   a. Strings and character manipulation,
   b. Arrays
   c. Files
   d. Struct data types.
2. Submission of the homework does not guarantee a grade, everyone is expected to defend verbally, or in a written way (such as Quiz) how, and why certain pieces of code (randomly selected by the instructor) used.
3. Any kind of act of plagiarism (borrowing from your friend, having homework is done by ChatGPT, or having someone do the homework for you etc.,) **is going to be dealt with formal misconduct procedures**.